SQL*Plus Setup &
Database Connection Guide for Mac

macOS • Oracle Autonomous Database • Wallet-based connection

Purpose
Install Oracle SQL*Plus on a Mac, configure the Autonomous Database wallet, and connect from Terminal.

Example service alias
`sqlworkshop_medium`

Before you begin
You need administrator/instructor-provided database credentials and wallet ZIP. Choose the Instant Client download that matches your Mac's processor.

Security reminder: A wallet is sensitive connection material. Keep it in a private folder and never publish it to GitHub or other public locations.

1. Identify your Mac and download Instant Client

Check the processor from the Apple menu → About This Mac. Apple Silicon Macs use Apple M-series chips; older Macs may use Intel. Download the matching macOS architecture from Oracle.

Mac type	Oracle download page	Packages required
Apple Silicon (M1/M2/M3/M4 and later)	Oracle Instant Client for macOS (ARM64)	Basic (or Basic Light) + SQL*Plus; use matching release versions.
Intel Mac	Oracle Instant Client for macOS (Intel x86)	Basic (or Basic Light) + SQL*Plus; use matching release versions.

Important: Download the Basic/Basic Light and SQL*Plus packages for the same architecture and release. Oracle's current macOS downloads provide DMG packages; follow the instructions shown on the matching Oracle download page.

2. Install the packages

1. Download and open the Basic (or Basic Light) DMG and the SQL*Plus DMG. Mount both packages in Finder.

2. Open Terminal. In Finder, open the mounted Instant Client volume and locate `install_ic.sh`.

3. In Terminal, change to the mounted volume containing the script. Example only—the volume name varies by version:

```
cd /Volumes/instantclient-basic-macos.arm64-23.xx.x.x.x
```

Replace the example volume path with the actual mounted volume name. You can type `cd` (including the space) and drag the mounted volume folder into Terminal to insert its path.

Run the installer script once, from one of the mounted Instant Client volumes:

```
sh ./install_ic.sh
```

Oracle's installer copies the mounted packages to a versioned folder under your Downloads directory, for example `~/Downloads/instantclient_23_26`. Use the actual folder created on your Mac. Eject the mounted DMGs in Finder when installation completes.

3. Add SQL*Plus to your Terminal PATH

The following assumes the installation folder is `~/Downloads/instantclient_23_26`. Replace it with your actual folder name.

```
echo 'export PATH="$HOME/Downloads/instantclient_23_26:$PATH"' >> ~/.zshrc
source ~/.zshrc
sqlplus -v
```

The final command should display the SQL*Plus version. macOS normally uses zsh as the default shell. If your Terminal uses another shell, add the PATH setting to that shell's startup file instead.

4. Configure the Autonomous Database wallet

1. Create a dedicated wallet folder in your home directory:

```
mkdir -p "$HOME/oracle/wallets/sqlworkshop"
```

1. Extract the wallet ZIP supplied by your instructor into this folder. Make sure files such as `tnsnames.ora` and `sqlnet.ora` are directly inside the folder, not nested inside another folder.

2. Open `sqlnet.ora` in a text editor and check the wallet directory setting. If it contains a placeholder path, replace it with the full path to the folder where the wallet files were extracted. Preserve the existing syntax and other settings.

```
Example wallet folder: /Users/your-mac-username/oracle/wallets/sqlworkshop
```

5. Set TNS_ADMIN permanently

Add the wallet folder to your zsh environment. This command appends the setting to `~/.zshrc`:

```
echo 'export TNS_ADMIN="$HOME/oracle/wallets/sqlworkshop"' >> ~/.zshrc
source ~/.zshrc
echo "$TNS_ADMIN"
```

The last command should print the wallet folder path. If you already have a TNS_ADMIN setting, edit the existing line rather than adding conflicting values.

6. Confirm the service alias

Open the wallet folder's `tnsnames.ora` file and check the exact alias. If it contains `sqlworkshop_medium`, use that spelling. Wallets may provide low, medium, and high service aliases.

7. Connect to the database

Open a new Terminal window and run:

```
sqlplus YOUR_USERNAME@sqlworkshop_medium
```

Replace `YOUR_USERNAME` with the username provided by your instructor. Enter the database password when prompted. Password characters may not appear as you type; this is normal. Do not put the password directly in the command.

8. Verify the connection

After SQL*Plus connects and displays the SQL> prompt, run:

```
SELECT USER FROM dual;
SELECT sys_context('USERENV', 'SERVICE_NAME') AS service_name
FROM dual;
```

The first query shows the database user. The second shows the service name for the session. To disconnect, enter:

```
EXIT
```

9. Common problems and fixes

Symptom / error	Checks and likely fix
zsh: command not found: sqlplus	Check that the PATH points to the actual Instant Client installation folder. Run source ~/.zshrc or open a new Terminal window, then try sqlplus -v.
ORA-12154: could not resolve connect identifier	Check that TNS_ADMIN points to the folder containing tnsnames.ora and that the alias is spelled exactly as listed.
ORA-12514 / ORA-12541 / timeout	Check the alias, database availability, network access, and whether the wallet belongs to the intended database.
ORA-01017: invalid username/password	Re-enter the username and password carefully. Ask the administrator to verify the account if the error continues.
Wallet or TLS error	Check the wallet directory in sqlnet.ora, the TNS_ADMIN value, and that the complete wallet ZIP was extracted.
Installer or library error	Confirm that the Basic and SQL*Plus packages match the Mac architecture and release, then follow the current Oracle instructions for that package.

10. Participant checklist

- ■ Downloaded matching Basic/Basic Light and SQL*Plus packages for the Mac's architecture.
- ■ Installed Instant Client using Oracle's installation instructions.
- ■ Added the actual Instant Client folder to PATH and verified with sqlplus -v.
- ■ Extracted the wallet ZIP into a dedicated folder.
- ■ Checked the wallet path in sqlnet.ora, where applicable.
- ■ Set TNS_ADMIN to the wallet folder and reopened Terminal or reloaded ~/.zshrc.
- ■ Confirmed the exact alias in tnsnames.ora.
- ■ Connected using sqlplus USERNAME@service_alias.
- ■ Successfully ran SELECT USER FROM dual;

Official Oracle resources

Instant Client for macOS (ARM64)
Instant Client for macOS (Intel x86)
Oracle Autonomous Database wallet instructions

Instructor note: Share the wallet through an approved secure channel and provide database usernames/passwords separately. The alias in this guide is an example; participants must use the alias in their own wallet's tnsnames.ora.